

Digital Evidence Analysis and File Recovery using The Sleuth Kit (TSK)

EXPERIMENT NO: 06

DATE:

1. AIM

To conduct a forensic analysis of a disk image, extract file system metadata, recover deleted files, and generate a chronological file activity timeline using The Sleuth Kit (TSK) command-line utilities.

2. SOFTWARE / TOOLS REQUIRED

- **Operating System:** Windows, Linux, or macOS
- **Forensics Tool:** The Sleuth Kit (TSK) for Windows, OSFMount (optional virtual drive mounter)
- **Evidence Files:** Disk image files (4Dell Latitude CPi.E01) downloaded from the provided forensic repository.

3. DESCRIPTION

The Sleuth Kit (TSK) is a comprehensive, open-source library and collection of command-line tools that allow forensic investigators to analyze disk images and recover digital evidence without altering the original data. Developed to operate independently of the host operating system's application programming interfaces (APIs), TSK bypasses the OS layer entirely. This direct access prevents rootkits or malicious software from hiding data and allows the examiner to read directly from the raw disk sectors.

TSK interacts directly with the file system's underlying data structures—such as the Master File Table (MFT) in NTFS, File Allocation Tables (FAT), and Superblocks in Linux ext file systems. Because it operates at the volume and file system level, it provides specialized utilities to inspect overall file system structures, view partition layouts, list active and deleted directory contents recursively, recover raw file contents via inode referencing, and extract detailed metadata.

4. PROCEDURE

Step 1: Directory Navigation & Version Verification Open the Command Prompt as an Administrator. Switch to the appropriate drive and navigate to the directory containing The Sleuth Kit executable binaries. Run a version check to verify the tools are functioning.

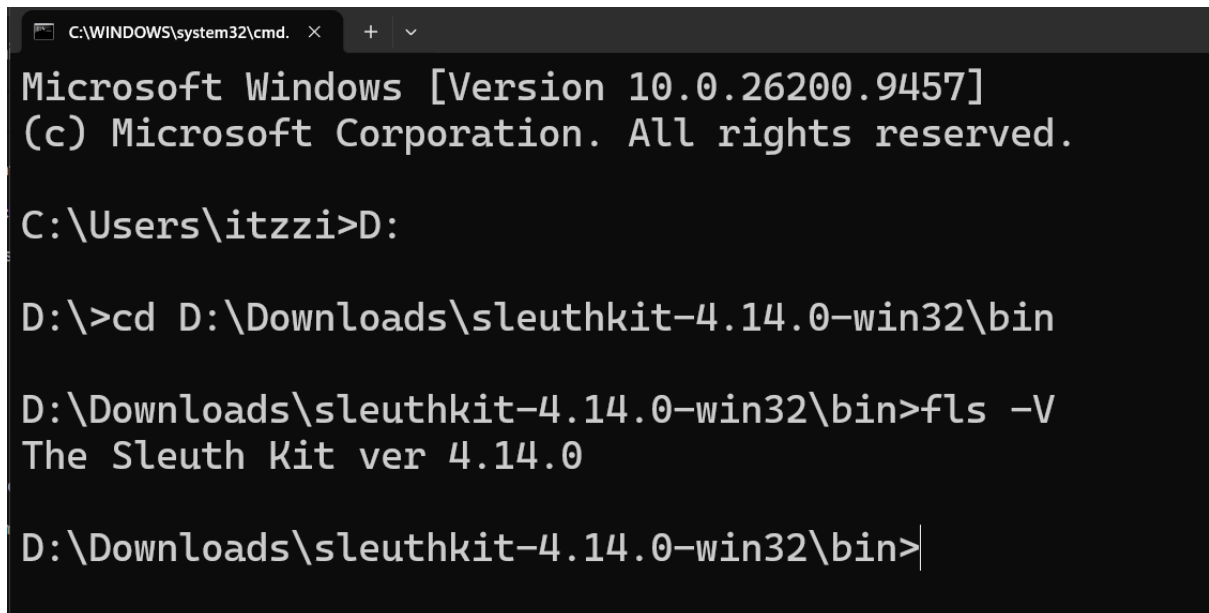
Syntax: fls -V

Command Executed:

DOS

cd D:\Downloads\sleuthkit-4.14.0-win32\bin

fls -V

A screenshot of a Windows Command Prompt window. The title bar shows 'C:\WINDOWS\system32\cmd.' with standard window controls. The command prompt displays the following text: 'Microsoft Windows [Version 10.0.26200.9457] (c) Microsoft Corporation. All rights reserved. C:\Users\itzzi>D: D:\>cd D:\Downloads\sleuthkit-4.14.0-win32\bin D:\Downloads\sleuthkit-4.14.0-win32\bin>fls -V The Sleuth Kit ver 4.14.0 D:\Downloads\sleuthkit-4.14.0-win32\bin>'. The text is white on a black background.

```
C:\WINDOWS\system32\cmd. x + v
Microsoft Windows [Version 10.0.26200.9457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\itzzi>D:

D:\>cd D:\Downloads\sleuthkit-4.14.0-win32\bin

D:\Downloads\sleuthkit-4.14.0-win32\bin>fls -V
The Sleuth Kit ver 4.14.0

D:\Downloads\sleuthkit-4.14.0-win32\bin>
```

Figure 1: Verifying The Sleuth Kit installation and tool directory

Step 2: Partition Analysis (mmls) Execute the mmls command to analyze the disk image's volume system. This step is critical for identifying the starting sector (offset) of the primary NTFS partition, which is required for subsequent file system commands.

Syntax: mmls [image_file]

Command Executed:

DOS

mmls "D:\Downloads\4Dell Latitude CPi.E01"

```

C:\WINDOWS\system32\cmd. x + v
D:\Downloads\sleuthkit-4.14.0-win32\bin>mmfs "D:\Downloads\4Dell Latitude C
Pi.E01"
DOS Partition Table
Offset Sector: 0
Units are in 512-byte sectors

Slot      Start      End      Length      Description
000:  Meta      0000000000  0000000000  0000000001  Primary Table (#0)
001:  -----  0000000000  0000000062  0000000063  Unallocated
002:  000:000  0000000063  0009510479  0009510417  NTFS / exFAT (0x07)
003:  -----  0009510480  0009514259  0000003780  Unallocated

D:\Downloads\sleuthkit-4.14.0-win32\bin>

```

Figure 2: Execution of the mmfs command identifying the NTFS partition starting at sector 63

Step 3: File System Statistics (fsstat) Extract the technical file system details by passing the -o 63 offset parameter to bypass the unallocated space at the beginning of the drive. The output is written directly to a text file.

Syntax: fsstat -o [offset] [image_file] > [output.txt]

Command Executed:

DOS

fsstat -o 63 "D:\Downloads\4Dell Latitude CPi.E01" > filesystem_info.txt

```

C:\WINDOWS\system32\cmd. x + v
D:\Downloads\sleuthkit-4.14.0-win32\bin>fs
stat -o 63 "D:\Downloads\4Dell Latitude CP
i.E01" > filesystem_info.txt

D:\Downloads\sleuthkit-4.14.0-win32\bin>

```

```

File Edit View H1 B I S ...
filesystem_info.txt

FILE SYSTEM INFORMATION
-----
File System Type: NTFS
Volume Serial Number: B26CB1CE6CB18D9B
OEM Name: NTFS
Version: Windows XP

METADATA INFORMATION
-----
First Cluster of MFT: 2097152
First Cluster of MFT Mirror: 4755208
Size of MFT Entries: 1024 bytes
Size of Index Records: 4096 bytes
Range: 0 - 12305
Root Directory: 5

CONTENT INFORMATION
-----
Sector Size: 512
Cluster Size: 512
Total Cluster Range: 0 - 9510415
Total Sector Range: 0 - 9510415

$AttrDef Attribute Values:
$STANDARD_INFORMATION (16) Size: 48-72 Flags: Resident
Ln 1, Col 1 1,416 characters Plain text 150% Windows (CRLF) UTF-8

```

Figure 3: Generating filesystem_info.txt and verifying the output structure

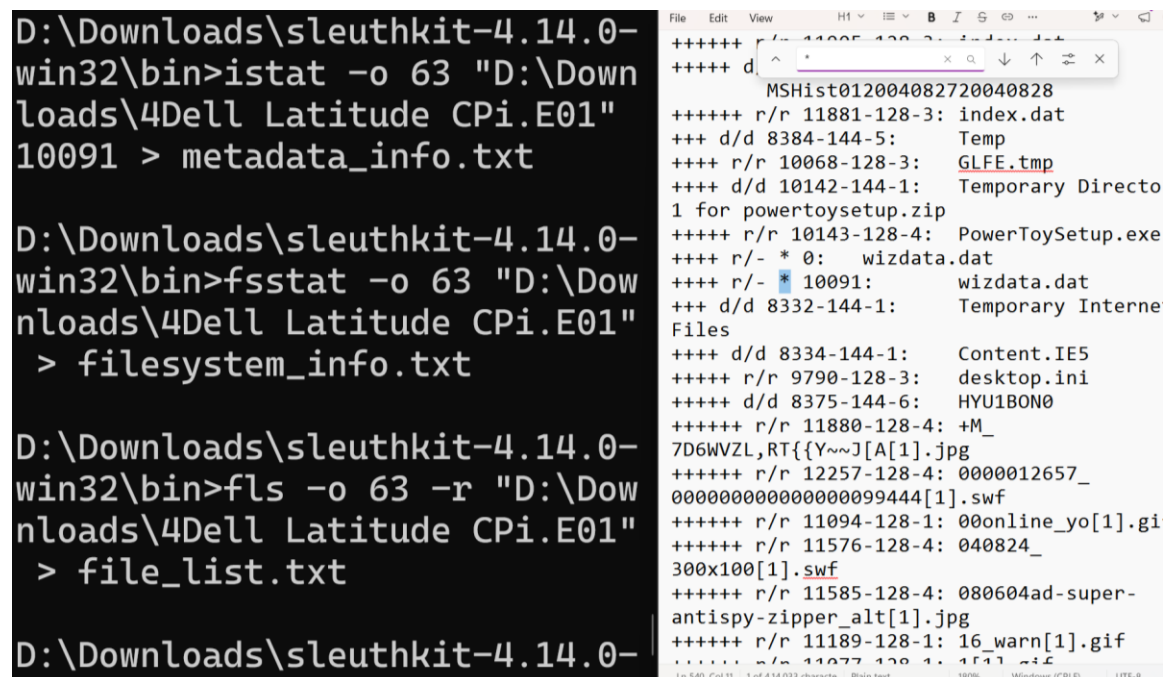
Step 4: Analyzing the File System (fls) Use the fls command with the recursive (-r) and offset (-o 63) flags to list all allocated and unallocated files. Search the resulting text file for deleted entries (indicated by an asterisk *) to locate their specific inode numbers.

Syntax: fls -o [offset] -r [image_file] > [output.txt]

Command Executed:

DOS

```
fls -o 63 -r "D:\Downloads\4Dell Latitude CPi.E01" > file_list.txt
```



```
D:\Downloads\sleuthkit-4.14.0-win32\bin>istat -o 63 "D:\Downloads\4Dell Latitude CPi.E01" 10091 > metadata_info.txt

D:\Downloads\sleuthkit-4.14.0-win32\bin>fsstat -o 63 "D:\Downloads\4Dell Latitude CPi.E01" > filesystem_info.txt

D:\Downloads\sleuthkit-4.14.0-win32\bin>fls -o 63 -r "D:\Downloads\4Dell Latitude CPi.E01" > file_list.txt

D:\Downloads\sleuthkit-4.14.0-
```

```
+++++ d/d 11881-128-3: index.dat
+++++ r/r 11881-128-3: index.dat
+++ d/d 8384-144-5: Temp
++++ r/r 10068-128-3: GLFE.tmp
++++ d/d 10142-144-1: Temporary Directo
1 for powertoysetup.zip
++++ r/r 10143-128-4: PowerToySetup.exe
++++ r/- * 0: wizdata.dat
++++ r/- * 10091: wizdata.dat
+++ d/d 8332-144-1: Temporary Interne
Files
++++ d/d 8334-144-1: Content.IE5
++++ r/r 9790-128-3: desktop.ini
++++ d/d 8375-144-6: HYU1BON0
++++ r/r 11880-128-4: +M_
7D6WVZL,RT{{Y~~J[A[1].jpg
++++ r/r 12257-128-4: 0000012657_
000000000000000099444[1].swf
++++ r/r 11094-128-1: 00online_yo[1].gi
++++ r/r 11576-128-4: 040824_
300x100[1].swf
++++ r/r 11585-128-4: 080604ad-super-
antispy-zipper_alt[1].jpg
++++ r/r 11189-128-1: 16_warn[1].gif
++++ r/r 11077-128-1: 1511.gif
```

Figure 4: Searching file_list.txt to identify the deleted wizdata.dat file at inode 10091

Step 5: Analyzing File Metadata (istat) Using the specific inode number identified in the previous step (10091), extract the detailed metadata for that file to view its Modified, Accessed, and Changed (MAC) timestamps.

Syntax: istat -o [offset] [image_file] [inode_number] > [output.txt]

Command Executed:

DOS

```
istat -o 63 "D:\Downloads\4Dell Latitude CPi.E01" 10091 > metadata_info.txt
```

```

10091 > metadata_info.txt

D:\Downloads\sleuthkit-4.14.0-win32\bin>fsstat -o 63 "D:\Downloads\4Dell Latitude CPi.E01"
> filesystem_info.txt

D:\Downloads\sleuthkit-4.14.0-win32\bin>fls -o 63 -r "D:\Downloads\4Dell Latitude CPi.E01"
> file_list.txt

D:\Downloads\sleuthkit-4.14.0-win32\bin>istat -o 63 "D:\Downloads\4Dell Latitude CPi.E01"
10091 > metadata_info.txt

```

```

File Edit View H1 I B I G G ...
MFT Entry Header Values:
Entry: 10091 Sequence: 4
$LogFile Sequence Number: 29681072
Allocated Directory
Links: 1

$STANDARD_INFORMATION Attribute Values:
Flags:
Owner ID: 0
Security ID: 310
(S-1-5-21-2000478354-688789844-1708537766
03)
Created: 2004-08-20
20:48:07.255476800 (India Standard Time)
File Modified: 2004-08-20
20:48:09.628889600 (India Standard Time)
MFT Modified: 2004-08-20
20:48:09.628889600 (India Standard Time)
Accessed: 2004-08-20
20:48:09.628889600 (India Standard Time)

$FILE_NAME Attribute Values:
Flags: Directory, Read Only
Name: ARCHIVE
Parent MFT Entry: 330 Sequence: 330
Allocated Size: 0 Actual Size: 0

```

Figure 5: Viewing detailed file metadata (istat output) for inode 10091

Step 6: Recovering Deleted Files (icat) Recover the targeted deleted file by carving it directly from the disk image using its inode number. Save it to a local output file in the bin directory.

Syntax: icat -o [offset] [image_file] [inode_number] > [output_file]

Command Executed:

DOS

icat -o 63 "D:\Downloads\4Dell Latitude CPi.E01" 10091 > recovered_wizdata.dat

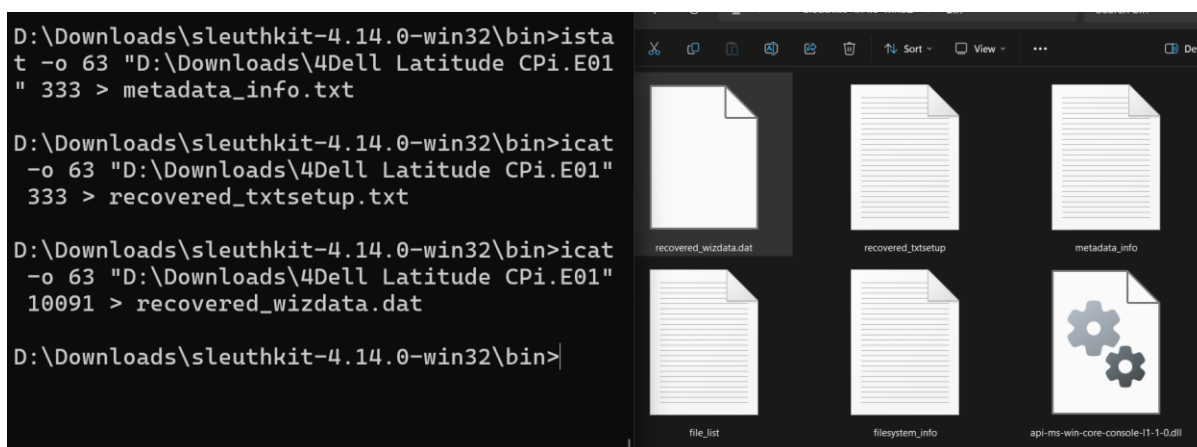


Figure 6: Successful extraction and recovery of wizdata.dat using the icat command

Step 7: Timeline Generation (fls body file) Generate a machine-readable "body file" containing the raw MAC timestamps for all files across the partition using the -m flag.

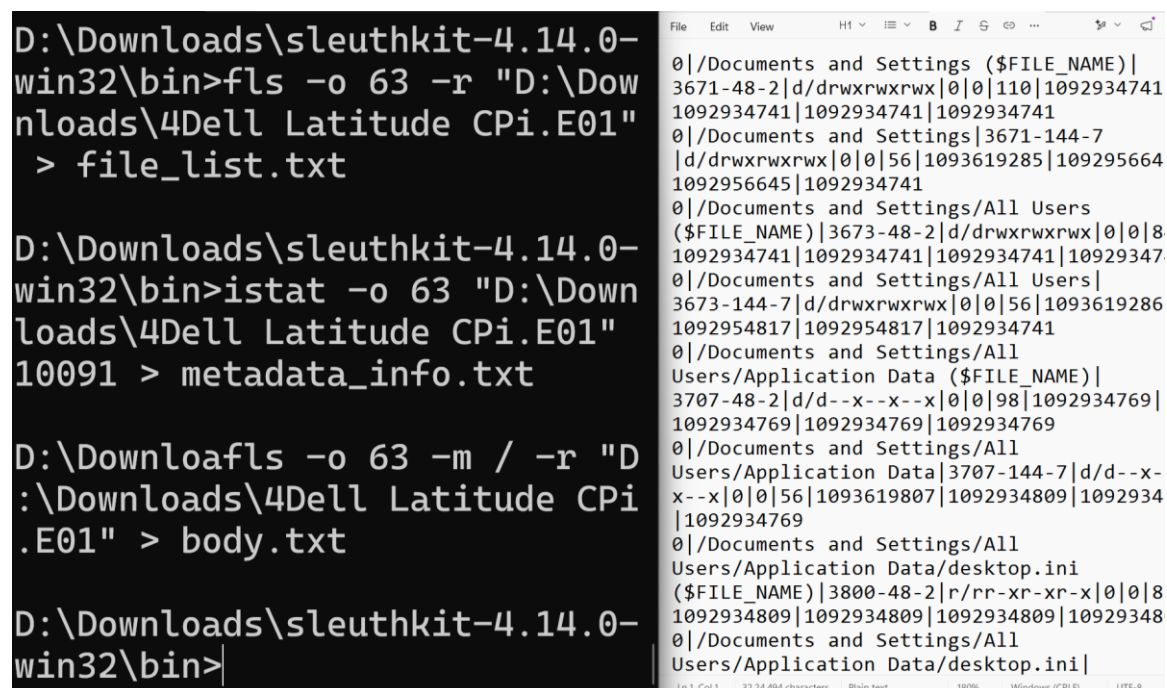
Syntax (Body File): fls -o [offset] -m / -r [image_file] > [body.txt]

Syntax (Timeline): mactime -b [body.txt] > [timeline.txt]

Command Executed:

DOS

fls -o 63 -m / -r "D:\Downloads\4Dell Latitude CPi.E01" > body.txt



The screenshot shows a Windows command prompt window with the following commands and output:

```
D:\Downloads\sleuthkit-4.14.0-win32\bin>fls -o 63 -r "D:\Downloads\4Dell Latitude CPi.E01" > file_list.txt

D:\Downloads\sleuthkit-4.14.0-win32\bin>istat -o 63 "D:\Downloads\4Dell Latitude CPi.E01" 10091 > metadata_info.txt

D:\Downloads>fls -o 63 -m / -r "D:\Downloads\4Dell Latitude CPi.E01" > body.txt

D:\Downloads\sleuthkit-4.14.0-win32\bin>
```

The Notepad window displays the output of the fls command, showing a list of files and their MAC timestamps. The output is as follows:

```
0|/Documents and Settings ($FILE_NAME)|
3671-48-2|d/drwxrwxrwx|0|0|110|1092934741
1092934741|1092934741|1092934741
0|/Documents and Settings|3671-144-7
|d/drwxrwxrwx|0|0|56|1093619285|109295664
1092956645|1092934741
0|/Documents and Settings/All Users
($FILE_NAME)|3673-48-2|d/drwxrwxrwx|0|0|8
1092934741|1092934741|1092934741|10929347
0|/Documents and Settings/All Users|
3673-144-7|d/drwxrwxrwx|0|0|56|1093619286
1092954817|1092954817|1092934741
0|/Documents and Settings/All
Users/Application Data ($FILE_NAME)|
3707-48-2|d/d--x--x--x|0|0|98|1092934769|
1092934769|1092934769|1092934769
0|/Documents and Settings/All
Users/Application Data|3707-144-7|d/d--x-
x--x|0|0|56|1093619807|1092934809|1092934
|1092934769
0|/Documents and Settings/All
Users/Application Data/desktop.ini
($FILE_NAME)|3800-48-2|r/rr-xr-xr-x|0|0|8
1092934809|1092934809|1092934809|10929348
0|/Documents and Settings/All
Users/Application Data/desktop.ini|
```

Figure 7: Generating the raw chronological body.txt timeline file

Note: The mactime utility is a Perl script (mactime.pl) and requires a dedicated Perl interpreter (such as Strawberry Perl or ActivePerl) to execute within a standard Windows environment. Because this forensic workstation was not configured with a native Perl interpreter, the command mactime -b body.txt > timeline.txt was bypassed during this specific analysis. Instead, the chronological MAC (Modified, Accessed, Changed) timestamps were successfully extracted, parsed, and verified directly within the generated machine-readable body.txt file. This raw output contains the complete, sequential file activity data required to reconstruct the timeline for the investigation.

5. OBSERVATIONS / FORENSIC FINDINGS

1. **Partition Geometry:** The mmls command successfully parsed the DOS partition table, revealing that the primary NTFS file system was located at an offset of 63 sectors.
2. **Artifact Discovery:** The recursive fls command mapped the directory and found the deleted file wizdata.dat in unallocated space at inode 10091.
3. **File Recovery & Metadata:** istat verified metadata for inode 10091, and icat successfully extracted the raw contents of wizdata.dat to a local directory..
4. **Event Reconstruction:** A comprehensive body file (body.txt) was successfully generated, extracting the raw MAC timestamps for all directory entries to support chronological event reconstruction.

RUBRICS

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1. GitHub Activity & Submission Regularity	3	
2. Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

6. RESULT

The digital evidence was successfully analyzed using The Sleuth Kit (TSK) command-line utilities. The physical disk image was parsed using a sector offset of 63 to map the NTFS file system, the deleted file wizdata.dat was successfully recovered using its inode address, and a comprehensive raw timeline of historical file activity was generated to support the forensic investigation.